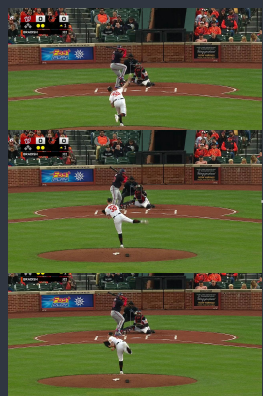


Video + Reference Construction

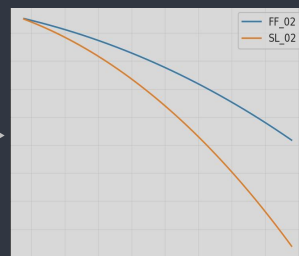


Pitch records

Pitch IDs
Reviewed 2D centers
Statcast fits

Frame/time matching

Reference-state construction
Effective camera estimation



Statcast-fitted 3D references

Reviewed video frames

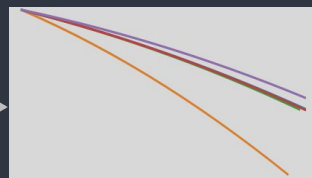
CONTROLLED GENERATION

Initial state + edit

6D initial state
FF / SL code
Optional velocity edit

Neural ODE rollout

Frozen BallODE
RK4 integration
Exact video timestamps



Predicted 3D trajectory

Fixed camera projection

K, R, T held constant

Screen-space control

Image centers
Nominal diameters
Motion direction



Dense RGB ball conditions

Shared scene + control

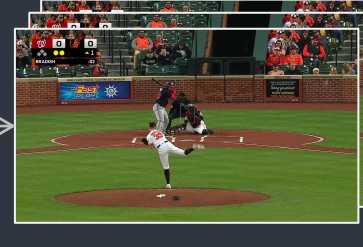
Same repaired background
Dense RGB conditions

Frozen video generator

16-frame RGB-conditioned video synthesis



Raw neural frames



Local composites

Physics-Guided Motion

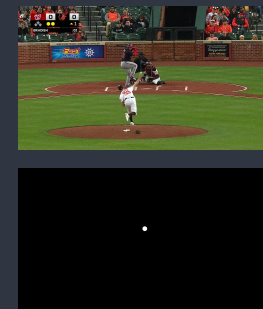
BallODE training

13 training pitches
Validation-selected weights

Neural acceleration

$[state_0, FF/SL_2]$
 $8 \rightarrow 32 \rightarrow 32 \rightarrow 3$
 $dp/dt = v$
 $dv/dt = g + a\theta$

Conditional Video Synthesis



Original frame & old-ball mask

ProPainter

Scene repair



Shared repaired scene

Pretrained video generator

AnimateDiff
+ SparseCtrl RGB
Weights frozen

frozen motion model

shared repaired scene

frozen weights